

Math 142, S14

Quiz 3

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Evaluate the integrals

(a) $\int \sec^4 \theta \, d\theta$. (Hint: use the substitution $u = \tan \theta$)

$$\begin{aligned}\int \sec^4 \theta \, d\theta &= \int \sec^2 \theta \sec^2 \theta \, d\theta \\ &= \int (1 + \tan^2 \theta) \sec^2 \theta \, d\theta \\ &\stackrel{u = \tan \theta}{=} \int 1 + u^2 \, du \\ &= u + \frac{u^3}{3} + C \\ &= \tan \theta + \frac{\tan^3 \theta}{3} + C\end{aligned}$$

(b) $\int_0^{\pi/2} \cos^2 \theta \, d\theta$. (Hint: use a half-angle identity)

$$\begin{aligned}\cos^2 \theta &= \frac{1}{2}(1 + \cos 2\theta) \\ \int_0^{\pi/2} \cos^2 \theta \, d\theta &= \int_0^{\pi/2} \frac{1}{2}(1 + \cos 2\theta) \, d\theta \\ &= \frac{1}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_{\theta=0}^{\theta=\pi/2} \\ &= \frac{1}{2} \left(\frac{\pi}{2} + \frac{1}{2} \cdot 0 \right) - \frac{1}{2} (0 + 0) \\ &= \frac{\pi}{4}\end{aligned}$$

Problem 2. (4 points) Find

$$\int \frac{1}{x^2 \sqrt{16-x^2}} dx$$

using the substitution $x = 4 \sin \theta$.

Let $x = 4 \sin \theta$, then $dx = 4 \cos \theta d\theta$.

$$\int \frac{1}{x^2 \sqrt{16-x^2}} dx = \int \frac{1}{(4 \sin \theta)^2 \sqrt{16-(4 \sin \theta)^2}} \cdot 4 \cos \theta d\theta$$

$$= \int \frac{1}{(16 \sin^2 \theta) \cdot (4 \cos \theta)} \cdot 4 \cos \theta d\theta$$

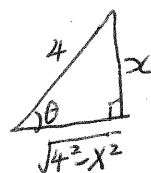
$$= \int \frac{1}{16 \sin^2 \theta} d\theta$$

$$= \int \frac{1}{16} \csc^2 \theta d\theta$$

$$= -\frac{1}{16} \cot \theta + C$$

$$= -\frac{1}{16} \frac{\sqrt{16-x^2}}{x} + C$$

$$\sin \theta = \frac{x}{4}$$



$$\Rightarrow \cot \theta = \frac{\sqrt{16-x^2}}{x}$$